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Course	FINS3645
Assessment	Project Part B – Stations 3 & 4 Data Factory Floor
Report Word Count	words
Live streamlit URL	https://z5481255projectb-a3qldyzumsumwzpmxqxxxn.streamlit.app/
Public repo link	https://github.com/emilijagliguroska/z5481255_projectB

Part B Overview

This report addresses Part B of the project, covering Station 3 and 4 of the Data Factory Floor (DFF) for the design and build of UNSW StableTrade, a prototype investment app offering several systematically managed funds. The product was built on structured data including 2020-2023 market prices on 50 US large-cap equities across ten sectors and 10 cryptocurrencies, as well as unstructured data from news headlines. StableTrade offers eight funds belonging to three portfolio types including equity-only, crypto-only, and combined equity-and-crypto. These portfolios were constructed according to four methods, namely, minimum-variance, maximum-Sharpe, risk-parity, and equal-weight. In Station 3, out-of-sample funds were built and backtested with no look-ahead. A new-sentiment index was constructed from headline data and integrated into the equity funds. Station 4 focused on implementing the product through a Streamlit app so investors can compare funds, read a fact sheet, set allocations, and monitor sentiment. Therefore, Part B delivers an investment app catered to the target user established in Part A and built on the data foundation.

1. The Funds and Backtest Design

1.1. UNSW StableTrade and its Target User

The app targets users who are risk-averse or pre-retirement and seek investments with exposure to both equity and crypto but also seek risk-parity weighting alongside management of the severity of portfolio drawdowns. StableTrade addresses the market gap of a systemically managed fund that follows a conservative risk framework, show through out-of-sample evidence. Each fund presents a unique return, volatility, and maximum drawdown, so an investor can monitor the risk they take on. Every fund follows explainable rules and current holdings are shown with the date they were last set. Minimum-variance and risk-parity funds are flagged as core options, leading risk-averse investors toward funds that manage crypto's upside while managing exposure to crypto's downside.

The data foundation preparation was preserved from Part A. The structured data included 50 US large-cap equities across ten sectors and 10 cryptocurrencies. Equities trade on a ~252-day calendar while crypto trade trades all 365 days per year. The unstructured data includes news headlines for the same 50 equities with duplicates removed based on matching ticker, date, and title, resulting in 146,836 unique headlines. All structured data spans 2 January 2020 to 29 December 2023 with timezone alignment before the data was merged.

1.2. Four Fund Types

Each fund combines a certain asset type and method. The four combined funds include Combined Min-Variance, Combined Max-Sharpe, Combined Risk Parity, and Combined Equal Weight, going beyond the required minimum of two methods. Equity Min-Variance and Equity Max-Sharpe were built using the 50 equities, along with Crypto Min-Variance and Crypto Max-Sharpe using the 10 cryptocurrencies. Two further versions of the Combined Min-Variance, Combined Max-Sharpe funds, enhanced with sentiment data, are discussed in Section 4. The optimiser only takes long positions, investing the full amount available,

and target weights are calculated using past data only ensuring the pipeline does not have access to future information.

1.3. Walk-Forward and Out-of-sample Backtesting

The backtest is walk-forward so that target weights are set using the prior one-year estimation window (252 equity trading days for combined and equity funds, 365 for crypto) at each rebalance, then applied from the day after the rebalance to the next. As such, investment decisions only rely on information already known from the day before. The out-of-sample period begins after the first rebalance occurs, once a full year of data exists. Out-of-sample trading first starts on 4 January 2021 for combined funds, 5 January 2021 for equity funds, and 2 January 2021 for crypto funds, running through to late December 2023. The Sharpe ratio assumes a zero risk-free rate and zero transaction costs. Returns are annualised using 252 days for equity/combined funds and 365 for crypto. Combined and equity funds rebalance every 21 trading days, and crypto funds every 30 equating to roughly once a month. Expected returns are derived from the historical average over the estimation window, and the covariance matrix employs an 0.05 intensity to prevent the solver failing on noisy data, falling back to equal weighting if the solution is invalid, while the risk parity uses the same shrunk covariance.

1.4. Calendars and Weekend Drop Decision

Crypto returns are calculated on the 365-day trading calendar, then merged onto the equity calendar with a left join. As the merge excludes crypto-only days, these days were dropped because the fund does not act on weekend-only price moves as it only trades on equity days. As a result, folding it into the next day would misattribute a move to a day it did not happen on. Also, the median share of each cryptocurrency's total return earned on the 454 crypto-only days was 21.5% (table 3), while for BCB it was 54.3%, and 137.5% for LTC. This means that weekend gains outweighed weekday losses, with BTC and ETH earning 5.1% and 14.1% of their return respectively on days equities were closed. Recalculating every crypto ticker's weekly return under both drop and fold was also tested. Across 2,090 ticker weeks, the two approaches gave opposite-signed returns 13.1% of the time, with a gap of around 2.4% per week. Specifically, BTC-USD rose 8.57% on Monday 30 March 2020, but had fallen a combined 8.6% over the prior weekend. Folding this in would have shown a loss instead of a gain. As folding assigns weekend moves to days they did not occur on, dropping these days was decided.

2. Out-of-sample Results and Fund Fact Sheets

2.1. The Combined Funds

Combined Min-Variance delivered a 5.8% annualised return, 12.6% volatility, a Sharpe ratio of 0.46, a maximum drawdown of 15.3%, and \$1 growing to \$1.18 over the out-of-sample period (table 1 and 2, figure 3). The largest weights at the last rebalance were all consumer and healthcare related (figure 3) being KO at 14.4%, WMT at 13.6%, and ABBV and MRK each at 9.5%. Combined Max-Sharpe performance stood out with a 25.0% annualised return, a Sharpe ratio of 1.01, and \$1 growing to \$1.95, but with nearly double the risk of the min-variance fund at 24.6% volatility and a 23.3% drawdown (table 1 and 2). It was also more concentrated, holding 46.3% in GE, 17.5% in NVDA, and 13.0% in BTC-USD at the last rebalance (figure 3). Figure 3 shows both funds side by side, with max-Sharpe providing more value during 2022. Combined Risk Parity sat in between (14.0% return, 16.1% volatility, Sharpe 0.87, drawdown 19.8%) with spread-out holdings and no position above 3.7%, while Combined Equal Weight returned 15.0% at 21.3% volatility with a Sharpe of 0.70. This ranking reflects the higher-risk funds being compensated for the deeper drawdowns they took on during 2021 to 2023.

2.2. Equity-only and Crypto-only Funds

Equity Min-Variance returned 5.0% at 12.7% volatility (Sharpe 0.40), and Equity Max-Sharpe returned 11.2% at 18.1% (Sharpe 0.62). Compared to the combined funds, adding crypto to the min-variance fund marginally changed its risk (12.6% vs 12.7%) while lifting its return (5.8% vs 5.0%) over 2021–2023. As such, crypto was diversifiable enough to add return without adding risk. Also, the combined max-Sharpe fund beat its equity-only version by a wide margin (1.01 vs 0.62 Sharpe) as it leaned heavily into crypto during a time crypto performed well. Crypto Min-Variance returned 38.1% annualised with 64.2% volatility

and a 73.9% drawdown. Crypto Max-Sharpe lost 14.3% annualised with 80.7% volatility, a negative Sharpe of -0.18, and an 88.5% drawdown, turning one dollar into 63 cents. This failure shows that with these ten assets, estimating expected returns from historical averages concentrates the portfolio onto a few coins, specifically, 68.7% in BTC-USD at the last rebalance, which was punished in the 2022 crash. This shows how optimisers can be fragile on a small set of assets, and it's why StableTradle leads with minimum-variance and risk-parity funds.

2.3. Out-of-sample Window Context

The three-year window covers more than one market. 2021 was a bull run for equities and crypto, 2022 was a shared downturn, and 2023 brought a partial recovery which was led mainly by large-cap equities. Figures 1 and 2 show where these periods appear in the fund paths so a user can see exactly how it performed during volatile periods.

3. The Sentiment Index

3.1. Scoring Headlines

Scoring headlines was based on the 146,836 de-duplicated headlines established in Part A. Every headline was scored with VADER which read the wording, punctuation, and tone of the sentence, returning a score between -1 and +1. The text was not cleaned before scoring since VADER relies on case and punctuation to judge tone correctly while smaller words change meaning depending on context, so they were maintained for accuracy. However, headlines are a rough measure of sentiment as plain VADER scored roughly half of finance headlines as neutral, meaning the model missed the signal. That gap is what motivated the custom word list added in Section 4.

3.2. Aggregation, No-Headline Policy, and Lag

Each headline gets its own score which is averaged across a stock's headlines for the day, and again averaged across all stocks in a sector with news that day. These sector scores are lined up against every equity trading day, including days with no news. On a no-news day, a sector simply keeps its last known score rather than being reset to zero, which would falsely suggest neutral news happened, or being skipped entirely, since even the quietest sectors rarely go more than a few days without news anyway. Finally, every score is shifted forward by one trading day, and this happens in exactly one place in the system. This means a trading decision only ever uses sentiment known as of the day before, so a headline from Saturday or Monday can't be used until Tuesday. Because the shift only happens once, there's no risk of it accidentally being applied twice or too early.

3.3. The Index

Figure 5 plots all ten sector lines over time. The scores stay close to zero, averaging about +0.13. Sectors tend to move together, since major news affects the whole market simultaneously, so the index is most useful for comparing a sector against its own normal range, which is how the app uses it. Second, cyclical sectors like Tech and Financials swing more than steadier ones like Utilities and Healthcare, matching expectations. Third, the index only reacts the day after a burst of headlines, confirming the lag is working as intended, while the flat stretches in the chart are quiet periods where the last score is simply carried forward.

4. Extensions and Innovation

4.1. Extending VADER

VADER is built for social media so finance-related words such could get accidentally scored as neutral. To fix this, the headline corpus was searched for finance words that appear often, are missing from VADER's list, and clearly point positive or negative. Each got a score on VADER's own scale, calibrated against similar words already in the list. Nothing existing was changed, so the extension only added signal. Loughran-McDonald dictionary words, originally implemented in Part A, were left out on purpose since they're built from filings rather than headlines and use a different scale. The impact was measured across the full corpus. The new words changed the score of 16,026 headlines (11%), lifting the average score by +0.015 per headline. Three words including beat, rise, and rally accounted for 77% of that shift, confirming the extension catches exactly the words the base model was missing. Every sector moved upward, since positive finance words simply appear more often in the data than negative ones.

4.2. Weekend Gap Analytics

The weekend-gap measurement used in part A to justify dropping weekend crypto returns is also a feature as it quantifies a risk that would otherwise be lost once the data is merged. Across the ten cryptocurrencies, the median share of total return earned on the 454 days equities were closed was 21.5%. This is shown on every crypto fund's fact sheet, so an investor can see how much of their return depends on weekend moves.

4.3. Sector-level Sentiment Fusion

The lagged sector sentiment score is standardised against the other sectors each day and every equity's weight is nudged using that score before being rescaled to sum to 1. Crypto isn't affected, since it has no sector. This stays safe from using future information since it relies only on an already-delayed score. It was applied to the two combined funds only, since they give the clearest before-and-after comparison. For Combined Min-Variance, this raised the Sharpe ratio from 0.457 to 0.489, lifted returns from 5.77% to 6.17%, and slightly reduced the worst drawdown which is a small but notable improvement with no added risk. For Combined Max-Sharpe, the effect was somewhat neutral. This is understandable as max-Sharpe is already concentrated in a few holdings, while the more spread-out min-variance is more impacted.

5. The Investor Journey in StableTrade

5.1. The StableTrade App

StableTrade is delivered as a Streamlit app, deployed from the public GitHub repository at (linked at top of report). It only reads precomputed results already saved to the repo including fund returns, weights, the sentiment index, and the various tables. Data loads once and is cached for 24 hours, so repeat visits are fast. The sidebar states the assumptions behind every number including zero risk-free rate, zero transaction costs, and the exact out-of-sample period used.

5.2. Four-Step Investor Journey

Step 1: Compare - The investor sees a full table of every fund's return, volatility, Sharpe ratio, drawdown, and growth of \$1, plus a chart comparing them, answering "which fund performed best or safest?".

Step 2: Fact sheet - Selecting a fund shows its key metrics, growth and drawdown charts, current holdings, and how it performed during three real stress periods (the 2022 downturn, the FTX collapse, and March 2023 stress). Crypto funds also show the weekend-gap table here, so that exposure is visible.

Step 3: Allocate - The investor sets weights across multiple funds using sliders, and the app builds and charts the resulting blended portfolio.

Step 4: Sentiment - The investor sees a risk gauge comparing each sector's current sentiment to its own normal range, can view any sector's sentiment over time, and sees the before-and-after impact of the sentiment tilt.

6. Critical Reflection

StableTrade's core design operates well as the no-look-ahead backtest was verified, the weekend-drop decision was backed by the 13.1% sign-flip rate and a clear reversal on BTC-USD, 30 March 2020, and the min-variance and risk-parity funds delivered the diversified, low-volatility profile the product promises. However, concentration in the max-Sharpe funds is a weakness with 46.3% in one stock (GE) combined and 68.7% in BTC-USD crypto-only. This has resulted from estimating returns from noisy averages over few assets, and the likely cause of the crypto fund's 88.5% drawdown. Three possible recommendations to improve the product include capping individual position sizes in the optimiser, adding transaction-cost penalty to the objective function to tests how much of max-Sharpe's edge survives real trading costs, and scaling the sentiment tilt strength to each portfolio's concentration rather than using a fixed 0.10. For the app, the Allocate tab lets an investor blend funds with sliders does not show how that specific blend would have performed during stress periods. While the Fact Sheet tab does this for individual funds, a custom allocation built by the user currently is not getting an equivalent stress-test view, which matters for the risk-averse investor StableTrade targets. Another improvement for the app is that current holdings are shown as a single snapshot with a date, but with no view of how those weights have shifted between rebalances, so adding a concentration-over-time chart could make the app more transparent and let investors track concentration risk.

Every fund was assessed through a walk-forward, out-of-sample backtest, using a one-year estimation window, no look-ahead, rebalancing every 21 trading days for combined and equity funds and every 30 for crypto, along with a zero risk-free rate and zero transaction costs. Testing on out-of-sample data, the combined maximum-Sharpe fund reached a Sharpe ratio of 1.01, while the minimum-variance fund reached 0.46 with the smallest drawdown of any fund, and the crypto funds displayed the high volatility expected of them. Separately, 146,836 news headlines were scored using VADER, strengthened with a 60-word finance lexicon developed from the headline corpus itself, then combined into a sector sentiment index that equally weights each headline and lags by one trading day. That index was then blended into the combined funds through a sector-level tilt, raising the minimum-variance fund's Sharpe ratio from 0.46 to 0.49 without adding risk, while making no meaningful difference to the maximum-Sharpe fund. A Streamlit app was built to let an investor compare funds, view individual fact sheets, construct an allocation, and check the sentiment analytics.

Appendix – Exhibits

Table 1: Out-of-sample Performance Metrics for all ten funds, row counts, first live date per fund, risk-free rate 0, zero transaction from 2020-01-01 to 2023-12-31.

Fund	Family	Method	Fused	First Live Date	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Growth of 1
Combined Min-Variance	Combined	min_variance	False	2021-01-04	0.05771	0.12631	0.45692	-0.15299	0.18254
Combined Max-Sharpe	Combined	max_sharpe	False	2021-01-04	0.24961	0.24627	1.01357	-0.23265	0.94613
Combined Risk Parity	Combined	risk_parity	False	2021-01-04	0.14011	0.16053	0.87281	-0.19782	0.47966
Combined Equal Weight	Combined	equal_weight	False	2021-01-04	0.14979	0.21252	0.70485	-0.28747	0.51753
Equity Min-Variance	Equity	min_variance	False	2021-01-05	0.05042	0.12650	0.39857	-0.15625	0.15811
Equity Max-Sharpe	Equity	max_sharpe	False	2021-01-05	0.11167	0.18148	0.61534	-0.24602	0.37151
Crypto Min-Variance	Crypto	min_variance	False	2021-01-02	0.38055	0.64209	0.59267	-0.73881	1.62889
Crypto Max-Sharpe	Crypto	max_sharpe	False	2021-01-02	-0.14259	0.80724	-0.17663	-0.88545	-0.36940
Combined Min-Variance + Sentiment	Combined	min_variance	True	2021-01-04	0.06173	0.12636	0.48856	-0.14744	0.19601
Combined Max-Sharpe + Sentiment	Combined	max_sharpe	True	2021-01-04	0.24999	0.24816	1.00740	-0.23635	0.94791

Table 2: Sentiment-tilt Fusion before vs after (tilt = 0.10 per cross-sectional z-score) from 2020-01-01 to 2023-12-31

Fund	Base Fund	Tilt	Base Ann. Return	Base Ann. Volatility	Base Sharpe	Base Max Drawdown	Fused Ann. Return	Fused Ann. Volatility	Fused Sharpe	Fused Max Drawdown	Sharpe Delta
Combined Min-Variance + Sentiment	Combined Min-Variance	0.1	0.05771	0.12631	0.45692	-0.15299	0.06173	0.12636	0.48856	-0.14744	0.03164
Combined Max-Sharpe + Sentiment	Combined Max-Sharpe	0.1	0.24961	0.24627	1.01357	-0.23265	0.24999	0.24816	1.00740	-0.23635	-0.00616

Table 3: Weekend-gap Analysis per Ticker, cum. return earned on the 454 crypto-only days (equities closed) vs the 1,006 equity-overlap days, Weekend Return pct = share of the full-period cumulative return earned on crypto-only days, from 2020-01-01 to 2023-12-31.

Ticker	Crypto Only Days	Equity Overlap Days	Crypto Only Cum. Return	Equity Overlap Cum. Return	Full Period Cum. Return	Weekend Return pct
BTC-USD	454	1006	25.0%	369.7%	487.0%	5.1%
ETH-USD	454	1006	232.1%	425.2%	1644.2%	14.1%
ADA-USD	454	1006	453.6%	220.8%	1676.0%	27.1%
BCH-USD	454	1006	14.6%	10.7%	26.8%	54.3%
EOS-USD	454	1006	-22.5%	-58.05%	-67.5%	33.3%
ETC-USD	454	1006	31.7%	269.4%	386.6%	8.2%
LTC-USD	454	1006	100.7%	-13.7%	73.3%	137.5%
TRX-USD	454	1006	114.2%	279.4%	712.6%	16.0%
XLM-USD	454	1006	57.7%	80.0%	183.7%	31.4%
XRP-USD	454	1006	22.4%	160.8%	219.2%	10.2%
Median	-	-	-	-	-	21.5%

Figure 1: Out-of-sample Growth of \$1 per combined-fund method (walk-forward, rebalanced every 21 trading day, OOS 2020-01-01 to 2023-12-31.

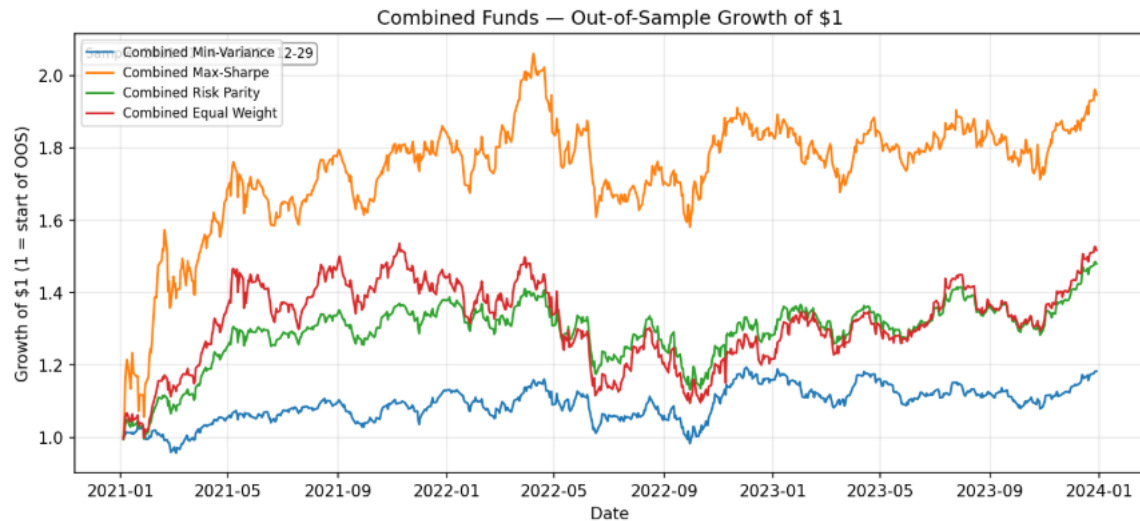


Figure 2: Out-of-sample Peak-to-Trough Drawdown for the combined min-variance and max-Sharpe funds, OOS 2020-01-01 to 2023-12-31

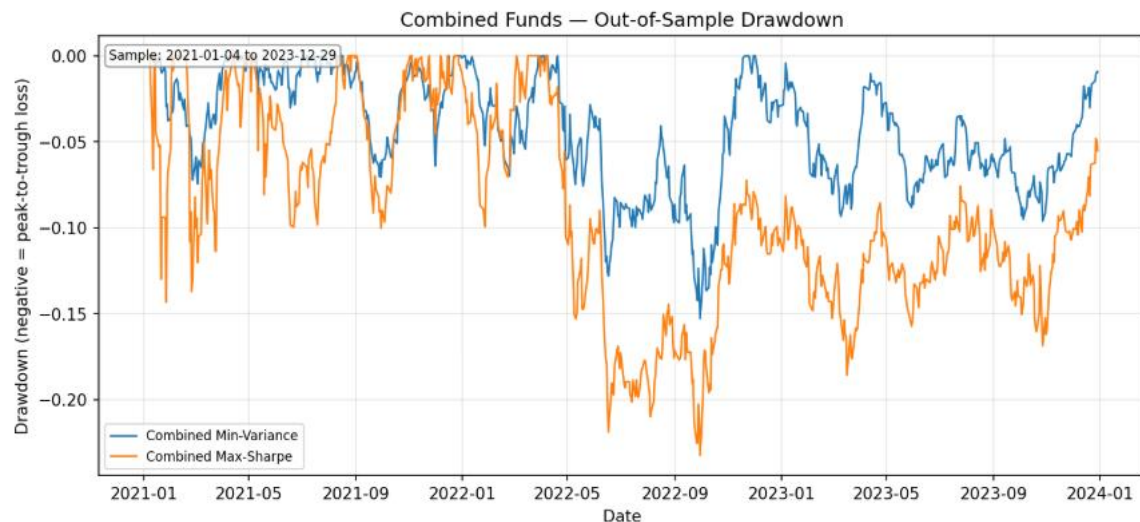


Figure 3: Sector Weights over time for the combined min-variance and max-Sharpe funds, crypto tickers aggregated into a single Crypto band, from 2020-01-01 to 2023-12-31

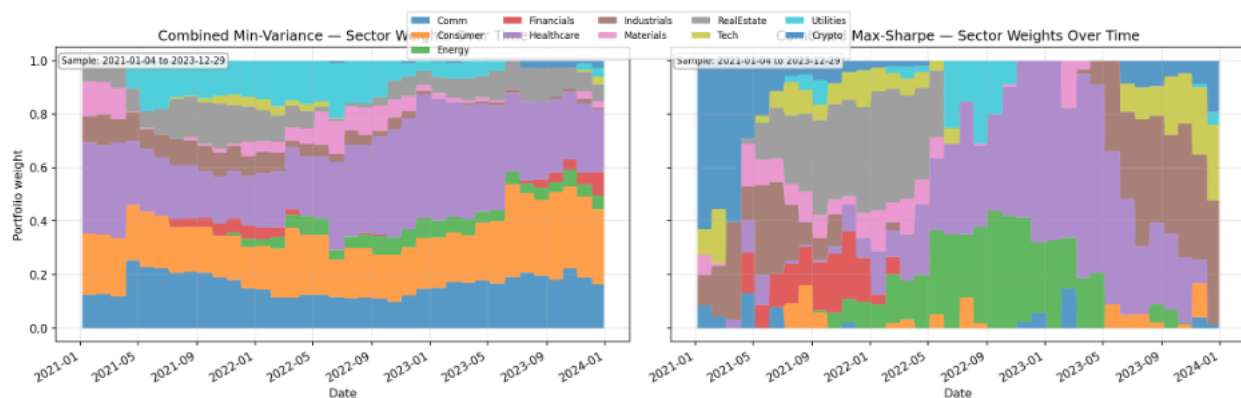


Figure 4: Annualised out-of-sample Sharpe Ratios (risk-free rate = 0) across funds and optimisation methods, OOS 2020-01-01 to 2023-12-31

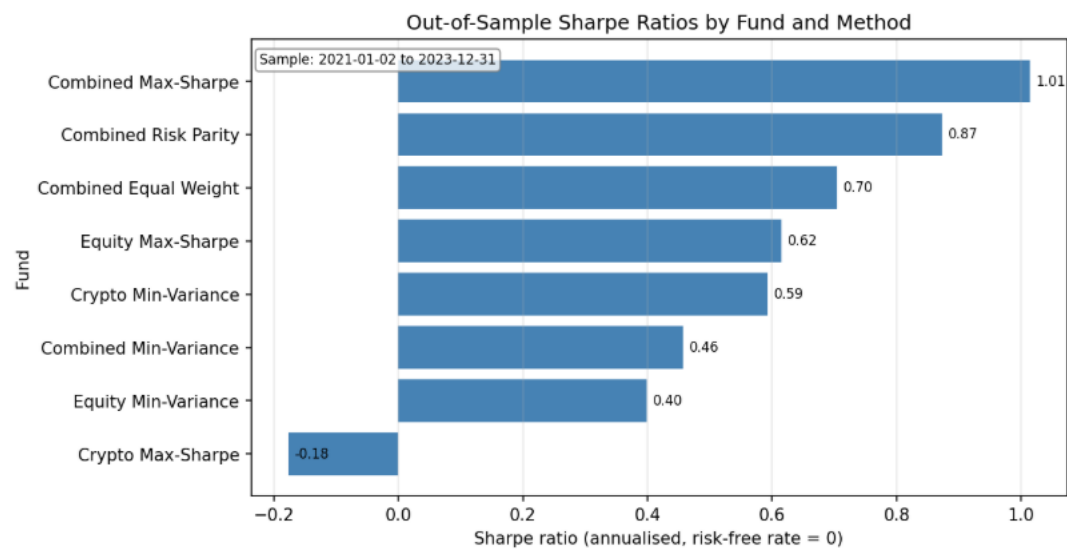


Figure 5: Equal-weighted sector sentiment index (VADER compound, carry-forward, one-trading-day lag, from 2020-01-01 to 2023-12-31

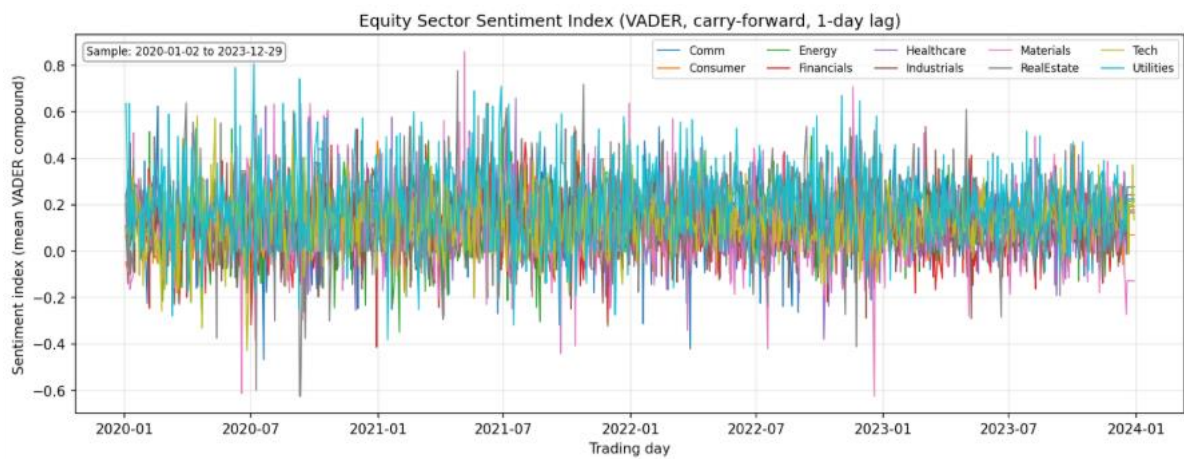


Figure 6: Sentiment-tilt fusion effect: combined min-variance fund before vs after the sector-level tilt (tilt = 0.10), 2020-01-01 to 2023-12-31

